

Technical documentation

Project: Sundown Studio — Landing Page Replica

Status: In Development (Refactoring Phase)

Author: K Madhu Satyakumar

Date: May 29, 2026

Version: 1.0.0

1. Executive Summary

The **Sundown Studio Landing Page Replica** is a high-fidelity, frontend-only recreation of the award-winning Sundown Studio website. The primary purpose of this project is to showcase advanced UI/UX interactions, complex CSS animations, high-performance scroll physics, and smooth canvas/vector-like layouts using only standard web technologies (HTML5, CSS3, Vanilla ES6+ JavaScript, and GSAP).

This project serves as a showcase of premium web engineering techniques, emphasizing micro-interactions, responsive styling, and modern organic aesthetics (e.g., gooey lava-lamp effects, cursor-tracking previews).

2. Product Vision & Core Objectives

- **1:1 Visual Fidelity:** Replicate the brand identity, typography, layouts, colors, and asset placements of the original Sundown Studio landing page.
 - **Buttery-Smooth Performance:** Maintain 60fps scrolling and animation cycles across modern desktop and mobile browsers.
 - **Zero-Build Architecture:** Rely entirely on vanilla files (HTML, CSS, JS) without a build step (no Webpack, Vite, or npm compilation required for execution), maximizing direct browser accessibility.
 - **Educational Showcase:** Document advanced animations (gooey morphs, smooth scroll, marquee loops, custom cursor trails) for portfolio review and development study.
-

3. User Experience & Design System

3.1 Typography

- **Primary Typeface:** Neue Haas Grotesk Text Pro (Regular, Medium, Bold) served locally via custom `@font-face` rules.
- **Backup Stack:** `sans-serif`, Arial.
- **Hierarchy:**
 - Loader Concept Words: `10vw`, Text-Transform: Uppercase.
 - Page 1 Hero Headline: `10vw` / `12vw` (highly responsive).
 - Section Titles: `1rem` + small orange circular bullet indicator.
 - Body Text: `1rem` to `1.2rem` with loose line-height (`1.3–1.5`) for premium editorial aesthetic.

3.2 Color Palette

- **Backgrounds:**
 - Primary light background: White/Off-white `#EFAE3`.
 - Secondary dark background: Deep Grey/Black `#0F0D0D`.
- **Primary Brand Color:** Bright Neon Orange (`#FE320A` / `#FE3F0A` for glowing elements).
- **Secondary Accents:** Soft Orange (`#FFE0D3`), Dark Grey (`#504A45` for inactive text).
- **Text:** Dark Charcoal (`#1C1B19`) on light backgrounds, Off-white (`#EFAE3`) on dark backgrounds.

3.3 Visual Style Elements

- **Goopy Morphs (Lava-Lamp):** Organic shapes utilizing heavy CSS filters (`blur(25px)` / `blur(30px)`) combined with a high-contrast container or layered transparency to create fluid merge-and-split visual effects.
- **Smooth Deceleration Curves:** Avoid default step-based browser scroll physics; integrate custom physics-based scroll decay.

4. Functional Requirements & Feature Specifications

F01: Preloader Screen (Intro Animation)

- **Description:** A cinematic intro overlay that loops through three core brand concepts before presenting the site.
- **Technical Specs:**
 - Fixed full-screen overlay with `#loader` (`z-index: 999`, `#000000` background).
 - Staggered animation cycle cycling through three headers: `ENVIRONMENTS` → `EXPERIENCES` → `CONTENT`.

- Staggering achieved via CSS keyframe animations and `animation-delay` offsets (1s, 2s, 3s).
- After 4 seconds, JavaScript changes the loader style (`top: -100%`) utilizing CSS transitions to slide the loader out of view.

F02: Navigation & Full-Screen Mobile Menu

- **Description:** Clean desktop header navigation that collapses into a custom full-screen slide-down menu overlay on tablet/mobile screens.
- **Technical Specs:**
 - Nav logo fetches the Sundown Studio vector logo dynamically.
 - Desktop: Displays links for `Work`, `Studio`, and `Contact` with smooth hover states.
 - Mobile ($\leq 600\text{px}$): Desktop links hide; a `#menu` button displays.
 - Clicking `#menu` toggles `#full-scr` (`top: 0` when open, `top: -100%` when closed) using smooth slide-down transitions.
 - The mobile menu overlay contains custom links and a pulsing background gooey blob.

F03: Hero Section & Autoplay Video

- **Description:** High-impact hero screen containing tagline, main bold headline, background gooey visual effects, and a background loop video.
- **Technical Specs:**
 - Columns auto-wrap on mobile (headline moves to top, tagline to bottom).
 - HTML5 video tag configured with `autoplay`, `loop`, `muted`, and `playsinline` (crucial for iOS/Safari autoplay compliance).
 - Background gooey blob `#hero-shape` with three child divs morphing via CSS keyframe loops and high blur parameters, sitting behind the video layer (`z-index` stack).

F04: Infinite Scrolling Marquee

- **Description:** Continuous, seamless, gap-free horizontal text loop displaying core offerings.
- **Technical Specs:**
 - Container `#move-text` set to `white-space: nowrap` and `overflow-x: hidden`.
 - Three duplicated `.con` text blocks loop horizontally.
 - Smooth looping uses CSS keyframes:

```
@keyframes move {
  from { transform: translateX(0); }
  to { transform: translateX(-100%); }
}
```

- Inter-word separator: `#gola` (a small orange circle bullet).

F05: Project List with Cursor-Tracking Preview

- **Description:** A list of recent studio work. Hovering over a project name slides up a hover overlay and reveals a floating image card that tracks the mouse cursor dynamically.
- **Technical Specs:**
 - Row hover event triggers `#overlay` CSS transition to slide orange block from `top: -100%` to `0%`.
 - Mouse movement over the container `#elem-container` tracks coords `dets.x` and `dets.y`.
 - GSAP `gsap.to()` smoothly moves the absolute-positioned `#fixed-image` div to match the cursor coords with a `0.4s` lag duration and `power3.out` easing curve.
 - Entering each project row `.elem` extracts its custom `data-image` URL attribute and swaps the `#fixed-image` background image dynamically.

F06: Interactive Services Tabs (Black Box Section)

- **Description:** A dark-themed container showcasing design services. Clicking tabs dynamically updates descriptive content and swaps the hero image.
- **Technical Specs:**
 - Tabs container (`#tabs`) contains clickable headers: `Design`, `Project`, and `Execution`.
 - Click event handlers swap active text styling (color, border-left accent).
 - The right-side image (`#page4-img`) is updated dynamically by reading the tab's `data-img` attribute.

F07: Brand Partnerships Swiper Carousel

- **Description:** A smooth, touch-draggable carousel displaying logos of partner brands with short contextual paragraphs.
- **Technical Specs:**
 - Swiper.js integration.
 - Configuration: `slidesPerView: "auto"`, `spaceBetween: 50`, `freeMode: true` (no rigid snapping, smooth slide deceleration), and `grabCursor: true`.
 - Displays partner SVG logos: Nike, Converse, Arc'teryx, Hunter, MediaLink, and Afterpay.

F08: Curtain-Reveal Footer

- **Description:** A cinematic footer that is slowly uncovered as the user scrolls to the bottom of the page.
- **Technical Specs:**
 - `#footer` is set to `position: fixed; bottom: 0; z-index: 9;`.
 - The main page wrapper `#main` has a relative position, background color, and `z-index: 10`.
 - An empty spacer section `#page5` is placed at the end of `#main`.
 - As `#main` scrolls up, the footer is revealed from underneath.
 - Background morphs in the footer use keyframed CSS blob shapes (`#shape1`, `#shape2`).

5. Technology Stack & Integration Details

Layer	Technology	Details / Configuration
Structure	HTML5	Semantic layout with <code>#main</code> section blocks, custom <code>data-*</code> attributes for dynamic content swapping.
Styling	CSS3	Custom <code>@font-face</code> loaders, CSS custom properties (variables), transitions, keyframe animations, <code>@media</code> queries for responsive layout adjustments.
Logic	ES6+ JavaScript	Vanilla DOM manipulation, event listener attachments, dynamic attribute handling.
Scroll Animation	Lenis.js	Physics-based inertial scroll setup. Configured with: <code>duration: 1.3</code> , <code>easing: 1.001 - 2^(-10t)</code> , <code>smoothTouch: false</code> (to preserve native mobile scroll feel).
Cursor Animation	GSAP 3.13	High-performance tweening library handles the trailing cursor effect for the hover image preview.
Slider Engine	Swiper 11	Powering the horizontal drag-to-scroll clients section.

6. Non-Functional & Performance Requirements

6.1 Performance & Rendering

- **Animation Smoothness:** Scroll interactions and gooey calculations must target **60fps** with no frame drops. Limit heavy filters on high-paint-area components where possible.
- **Asset Loading:** The hero video asset (~16MB) should load asynchronously and play in compliance with mobile hardware configurations. In future releases, video and WebP compression must be prioritized to keep page-load times under **2.5 seconds** on a 3G network connection.

6.2 Responsiveness & Adaptability

- **Breakpoints:**
 - Mobile: **<= 600px** (Adjusts grid column layout, triggers sticky slide overlay menu, scales down text headers).
 - Tablet: **601px - 1023px**.
 - Desktop / Ultrawide: **>= 1024px**.
- Touch-handling overrides: Lenis scroll easing disabled on touch interfaces to allow native OS-level scrolling.

6.3 Browser Compatibility

- Must function identically across all modern rendering engines: Chromium (Chrome, Edge, Opera), WebKit (Safari), and Gecko (Firefox).

7. Future Roadmap & Enhancement Backlog

- [] **Accessibility (a11y) Audit:** Add **aria-*** tags, keyboard navigation capabilities to service tabs and client sliders, and proper alt text on all icons.
- [] **Asset Optimization:** Compress hero video format (use WebM with MP4 fallback) and scale responsive WebP images via **srcset** tags.
- [] **State-Driven Tabs:** Refactor service tabs to load descriptions and metadata from a clean JSON config file rather than hardcoding long text directly into the DOM/JS functions.
- [] **Clean Code Refactoring:** Transition style definitions out of inline selectors and consolidate CSS files; clean up unused desktop/mobile duplicate elements (such as duplicate **#full-scr** structures).